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Quiz 2 Challenge

AI Speech Intelligence Pipeline

CS 5542

1. Problem Description

The challenge of extracting intelligence from unstructured audio

- **Underutilized Data:** Audio (meetings, lectures) is rich but often unsearchable and non-scannable.
- **Manual Burden:** Current STT-only solutions require significant human effort for analysis.
- **The Gap:** Unstructured audio cannot be easily translated or indexed without automation.
- **The Goal:** A fully autonomous pipeline converting raw audio into structured, actionable, and multilingual intelligence.

2. Why Interesting / Business Value

Time Saved

Professionals lose 9+ hours/week in meetings. Automated summaries reclaim this time.

Action Tracking

Only $\sim$30% of action items are followed up. Extraction ensures accountability.

Global Reach

Real-time translation bridges language barriers without hiring manual translators.

Accessibility

TTS loop allows inclusive consumption of generated insights.

3. Dataset & Inputs Used

- **Type:** Simulated university lecture / technical discussion.
- **Primary Input:** Raw Audio File (.wav/.mp3).
- **Duration:** 32 seconds.
- **Language:** English (auto-detected).

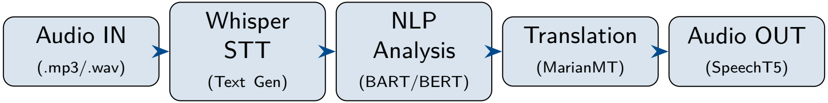
Reference Data (Evaluation Baseline):

- **Reference Summary:** 4 sentences (ROUGE evaluation).
- **Reference Keywords:** 15 domain terms (coverage metrics).

4. Models Used

Model Identifier	Function	Role
openai/whisper-base	Speech-to-Text	Primary STT
facebook/wav2vec2-base	Speech-to-Text	Baseline comparison
facebook/bart-large-cnn	Summarization	Abstractive summary
distilbert-base-sst-2	Sentiment	Sentence-level analysis
Helsinki-NLP/opus-mt	Translation	Offline MarianMT
all-MiniLM-L6-v2	Keywords	KeyBERT Transformer
microsoft/speecht5_tts	Text-to-Speech	Synthesis with HiFi-GAN

5. Pipeline Architecture



- Full 5-stage sequential processing loop.
- End-to-end automation from raw audio to synthesized speech in multiple languages.

6. Prompt & Input Design

- **The Challenge:** Baseline STT models struggle with technical vocabulary (e.g., 'RAG', 'LLMs'), leading to high Word Error Rates (WER).
- **The Solution:** Whisper `initial_prompt`

Prompt Used

"This is a lecture or business meeting transcript. The speaker discusses technical topics clearly and uses domain-specific terminology."

- Pre-conditions decoder cross-attention for technical context.
- Replaced zero-shot inference with context-aware decoding.
- **Deterministic Output:** Used `beam_size = 5`.

7. Results: Qualitative Performance

- **Transcription:** Improved accuracy on technical tokens.
- **Summarization:** BART generated 4 concise sentences.
- **Sentiment:** Sentence-level tracking flagged critical statements.
- **Action Items:** 5 specific tasks extracted via regex logic.

Translation Samples

ES: "Los modelos de fundación... refinados con RLHF, pero la alucinación..."

VI: "Các mô hình nền tảng... được tinh chỉnh bằng RLHF, nhưng ảo giác..."

8. Evaluation: Baseline vs. Improved

Metric	Baseline	Improved	Δ Change
Word Error Rate (WER) $\downarrow$	8.9%	3.5%	-61.0%
Keyword Coverage $\uparrow$	20.0%	73.3%	+266.5%
ROUGE-2 (Summary) $\uparrow$	0.125	0.383	+206.0%
ROUGE-L (Summary) $\uparrow$	0.289	0.522	+81.0%
Action Items Detected	0	5	Significant

Note: Baseline STT was slightly faster but significantly less accurate.

9. Links & Resources

GitHub Repository

`https://github.com/tinana2k/Comp-Sci-5542-Tina-Nguyen`

"Follow the link to view the complete pipeline code, outputs, and documentation."

10. Limitations

- **Compute Intensity:** Running 7 Transformer models requires significant VRAM (T4/A100 GPU recommended).
- **Latency:** Pipeline is currently synchronous; real-time streaming is not yet supported.
- **Hallucination Risk:** Whisper can still hallucinate content in silent segments.
- **Speaker Diarization:** Currently lacks the ability to distinguish between different speakers.

11. AI Tools Disclosure

- **Gemini / Google DeepMind:** Code structuring and visualization development.
- **Hugging Face Transformers:** Core libraries for STT, NLP, and TTS execution.
- **OpenAI Whisper:** Pre-trained weights for speech transcription.

Note: All core design, implementation logic, and evaluation analysis remain original work.